

A Conserved Clock Rate, and the Board Run End to End

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Abstract

The solar system demands a clock running $s_0 \gtrsim 1.5 \times 10^7$ times proper time [8], a number nothing explained. We show it does not need explaining. The closure's two kinetic coefficients descend from a single logarithmic function $P(X) = -\frac{U}{2} \log(U - 2dX)$ whose pole is a *speed limit*, with $1/m_{\text{rel}}$ its Lorentz factor; and since every term of the action carries χ only through derivatives, the shift symmetry is exact and that Lorentz factor is its conserved Noether charge. The clock rate is therefore set once and protected, which makes a large value technically natural rather than tuned. The matter coupling MOND requires breaks that symmetry, driving the charge linearly in cosmic time and, through $a_0 = \lambda^3/(12\pi G\beta s_0)$, driving a_0 down; the breaking rate is independent of s_0 , so holding a_0 flat to one percent since $z = 5$ forces the sector's equation of state below 5.7×10^{-7} , more than two orders under the acoustic bound. Criticality survives there — its growth beats dilution seventeenfold at the Lyman- α epoch — and supplies an exact *operating condition*, $w > 2/\kappa^2$, twice the residual sound speed the same mechanism ends at. We then run every gate at one parameter point. Two gates fix two numbers and the rest follow; all twelve pass simultaneously; the four at unit margin are exactly the four parameters placed on their boundaries on purpose; and two combinations derived in different lanes from different physics agree to one part in 10^7 with the residual exactly $s_0/(s_0 - 1)$. The run also exposes a coupling not previously stated: the flat- a_0 gate constrains the *product* Cw , so the depth of the MOND interpolation and the equation of state trade off, and honouring that the allowed set has a genuine interior with every gate clearing by at least a factor of two. Seven quantities are fixed by a gate or a relation and five remain free, two bounded. We also report, in full, two intermediate results of our own that later lanes overturned — a claimed naturalness pincer and the ultraviolet cutoff it rested on — together with the audit that overturned them. **This is a construction closed on its own gates, not a complete theory:** every gate is an analytic estimate rather than a likelihood, no Boltzmann, N -body or loop calculation exists, and κ remains provably underivable by this action class. 147 Lean 4 theorems, zero **sorry**.

1 The clock rate is a conserved charge

The action [2] carries χ only through derivatives: $P(X, \tau)$ with $X = Q^2 - Y$, $sW(Y, \tau)$, and $\gamma X \square \chi$. The shift symmetry $\chi \rightarrow \chi + c$ is therefore exact, and a global shift symmetry of a scalar has no anomaly. Its Noether charge is $Q_\chi = a^3 P_X \bar{q}$.

Separately, the closure's coefficients $P_X = Ud/m$ and $P_{XX} = 2Ud^2/m^2$, with $m = U - 2dX$, are *not* independent choices: both follow from

$$P(X) = -\frac{U}{2} \log(U - 2dX), \tag{1}$$

whose pole at $m = 0$ sits at $X_* = U/(2d)$ where $P_X \rightarrow \infty$. Infinite energy is required to reach it, so the field can approach that invariant and never cross: it is a *speed limit*, and $1/m_{\text{rel}} = U/(U - 2dX)$ is exactly its Lorentz factor. On the derived family $Q_\chi = \bar{q}d/m_{\text{rel}}$, constant, so with the clock identity $s_0 - 1 = w/m_{\text{rel}}$ [6],

$$\boxed{s_0 - 1 = \frac{w Q_\chi}{\bar{q} d}} \quad (2)$$

A clock rate of 10^7 is the field sitting close to its own speed limit, with the distance fixed by a conserved charge — set once and protected, which is what makes a large value technically natural. Nothing needs to explain it epoch by epoch.

A correction to [7] falls out. In the limit the decoupling locus forces, $2d\bar{q}^2 = U$, the enhancement ratio is $C = 2k\bar{q}^2/U = k/d$, which is exactly the reciprocal of the interpolating function's floor. The inequality $C \gtrsim 10^3$ reported there as the price of putting a force law on the locus *is* the rotation-curve requirement restated, and is withdrawn as a separate condition.

2 What MOND's coupling costs

$\lambda\chi\rho_b$ is not shift symmetric: $\partial_t(a^3P_X\bar{q}) = \lambda a^3\rho_b$, a constant, so the charge grows linearly in cosmic time and s_0 with it. Since $a_0 = \lambda^3/(12\pi G\beta s_0)$ with λ, β constants of the action, a growing clock rate pulls a_0 *down*. Solving the solar-system condition $f_s = 1$ for the coupling and substituting the branch relations,

$$(\lambda\bar{q})^2 = 4\pi CGw\rho, \quad \frac{\partial(\lambda\bar{q})^2}{\partial s_0} = 0, \quad (3)$$

so the clock rate cancels and the drift cannot be tuned away. With $G = 1/(8\pi M^2)$ and $\rho = 3H^2 M^2 \Omega$, $\lambda\bar{q} = H\sqrt{3Cw\Omega/2}$ and the charge grows at $2\sqrt{1.5Cw\Omega_{\text{dm}}}(\Omega_b/\Omega_{\text{dm}}) = 0.074$ per e-fold, so a_0 falls 12.4% since $z = 5$ against the one percent the derived flat law permits. Holding it forces

$$Cw \leq 5.7 \times 10^{-4}, \quad \text{hence} \quad w \lesssim 5.7 \times 10^{-7} \text{ at } C = 10^3, \quad (4)$$

177 $\times$ tighter than the acoustic bound and the tightest constraint on the sector's equation of state this programme has derived. The published prediction band $0 < w \lesssim 10^{-4}$ shrinks by that factor; its *sign* survives, and with it the distinction from a cold sector's exact zero, but its magnitude moves further out of reach.

3 Criticality there, and an operating condition

Every gate from [4, 5] was run at $w = 10^{-4}$. Setting the criticality growth term equal to the dilution term at zero gradient, where the driver is weakest, gives

$$w > \frac{2}{\kappa^2} = 2 \times (\text{residual } c_s^2), \quad (5)$$

the equation of state must exceed twice the residual sound speed the same mechanism ends at. At $w = 5.7 \times 10^{-7}$ the growth beats the dilution seventeenfold at the Lyman- α epoch and the mechanism runs from $z \approx 940$ down, covering the forest, galaxy formation, lensing and S_8 . Demanding it at recombination instead is not met, and is not required: the sound-speed correction to the sector's perturbation equation at the third-peak scale is 4.9×10^{-4} .

4 Two results of ours that were overturned, and by what

We record these in full because the programme’s rule is that a withdrawn result is reported as loudly as a standing one.

Withdrawn: a naturalness pincer. We estimated the radiative correction to the clock coefficient as $\delta U = \Lambda^4/(16\pi^2)$ with Λ the cutoff, concluded $\delta U/U = \sqrt{2/m_{\text{rel}}}/8 \approx 9 \times 10^5$, and reported a twelve-order pincer between radiative stability and the solar system, with no interior. **This is wrong.** The scalar’s dispersion is $\omega = c_s k$, so the mode sum carries a factor c_s , while the cutoff carries one inverse power of it: the estimate was too large by exactly $1/c_s$.

Withdrawn: the cutoff number. We reported a cutoff of 2.3 mm, $\kappa = 1.3 \times 10^{28}$, from the field-space distance to the speed limit. The sound speed used there was computed from $P(X)$ alone, dropping the cusciton’s own gradient contribution; the full result [3] carries a factor $(1 - s_0)$, a different sign and seven orders, and at zero gradient the physical sound speed is imaginary. **The number should not be quoted until recomputed.**

And a caution about the correction itself. That $\delta U/U = 1/8$ comes out identical at sound speeds spanning twenty-two orders is not robustness. U is the only dimensionful coefficient and the cutoff was derived from U , so $\delta U \propto U$ is forced by dimensions; the content is the pure number, which such an estimate fixes to about an order. The honest statement is $\delta U/U = O(1)$: marginally stable.

What survives, and it needs no loop estimate. The conserved charge gives $m_{\text{rel}} = a^3 \sqrt{dU/2}/Q_\chi$, so a small margin *is* a large charge and there is no difference of large numbers anywhere. The sensitivity is a square root, $d \log m_{\text{rel}}/d \log U = \frac{1}{2}$, so the margin stops being small only if $\delta U/U \geq 2/m_{\text{rel}} = 5.3 \times 10^{13}$ — eight orders beyond even our own overstated estimate. The margin is safe on either arithmetic, and the pincer was an error of comparison, not of estimation.

5 The board, end to end

Two gates fix two numbers — the flat- a_0 drift fixes $w = 5.66 \times 10^{-7}$, the alignment fixes $s_0 = 1.5 \times 10^7$ — and $m_{\text{rel}} = 3.77 \times 10^{-14}$, $U = 3.68 \times 10^{-25} \text{ eV}^4$ and $\mu = 1 - m_{\text{rel}}$ follow.

gate	measured	threshold	margin
acoustic scale	5.66×10^{-7}	10^{-4}	177
flat a_0 drift, $z \leq 5$	1.00×10^{-2}	10^{-2}	1.0
criticality operates	5.66×10^{-7}	1.95×10^{-9}	290
forest residual	9.77×10^{-10}	10^{-9}	1.02
positivity of the sector	1.5×10^7	2	7.5×10^6
margin below equation of state	3.77×10^{-14}	5.66×10^{-7}	1.5×10^7
cubic backreaction	2.94×10^{-9}	0.044	1.5×10^7
preferred-frame mixing	0	10^{-4}	∞
$\gamma_{\text{PPN}} - 1$	0	2.3×10^{-5}	∞
alignment / α_1	1.0	1.0	1.0
radiative stability	$O(1)$	5.3×10^{13}	5.3×10^{13}
MOND floor = window	10^{-3}	10^{-3}	1.0

All twelve pass simultaneously, which no previous result established. The four at unit margin are exactly the four parameters placed on their boundaries on purpose, so the board carries no hidden over-constraint. And two combinations derived in different lanes from different physics had to agree

and do: $s_0 U = 5.5196 \times 10^{-18}$ from the force-law matching against $w\rho = 5.5196 \times 10^{-18}$ from the symmetry-breaking drift, ratio 1.00000007, the residual exactly $s_0/(s_0 - 1)$.

A coupling the run exposed. Moving all four parameters off their limits naively fails the flat- a_0 gate, because the drift goes as $\sqrt{C}w$: **that gate constrains the product Cw , not w alone**, so the depth of the MOND interpolation and the sector's equation of state trade off. Honouring $Cw \leq 5.7 \times 10^{-4}$ — at $w = 5 \times 10^{-8}$, $s_0 = 10^8$, $C = 3 \times 10^3$, $\kappa = 10^5$ — every gate clears with smallest margin 1.94. The allowed set is a region, not a corner.

The count. Seven quantities are fixed: the family exponents by w ; γ by $W_0 = 0$; d by $\mu \rightarrow 1$; the U amplitude by ρ_{dm} ; m_{rel} by w and s_0 ; λ^3/β by a_0 on either footing; and k by $f_s = 1$. Five remain free: w and s_0 , both bounded by gates; a λ - β split no observable sees; ℓ , entering one inequality; and κ .

6 What is not claimed

This is not a complete theory of gravity and nothing here should be read as one. Every gate above is an analytic result inheriting its own lane's limits. **There is no Boltzmann calculation with this sector present, no N -body run, no loop computed anywhere, and no likelihood:** the board says the construction is not obviously inconsistent, not that it fits the data. The reach κ is a deliberately conservative stand-in because our own cutoff was withdrawn; recomputing it moves the criticality floor down only. The health condition $U > 4d\ell$ is not evaluated because ℓ has never been pinned. The disformal coupling's effect on the cosmological sector is not computed. The drift is linear order with a conformal coupling to baryons alone. The one percent flat-law tolerance is taken from the framework's own derived law, not from data. And $\kappa = \frac{1}{2}$ remains fitted and *provably underivable by this class of actions* [1] — a theorem of this programme, not a gap in this paper, and the reason a complete theory is not available on this action however long one pushes.

7 Reproducibility

fable_independent_2026/ lanes L217–L223 with their committed outputs and results files (commits e1c8071de, 7445dbcbc, 5958ddeff, 974cf087b, 081544dac, f8dd67953, aeae2b650). The Lean file fable_independent_2026/lean_2026/Mondlean.lean: 147 theorems, exit 0, zero sorry, axioms $\subseteq$ {propext, Classical.choice, Quot.sound}.

References

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